

## **CURRICULUM VITAE**

7/2/2026

Philip J. Freda, Ph.D., M.S.

### **CURRENT POSITIONS:**

2025-present                      Research Scientist I, Department of Computational Biomedicine, Cedars-Sinai

### **PROFESSIONAL CONTACT INFORMATION**

Business Address: Cedars-Sinai Medical Center, Computational Biomedicine, 700 N. San Vicente Blvd., Pacific Design Center, Suite G540, West Hollywood, CA, 90069, United States

Phone: 310-423-6573 (Office), 215-439-8545 (Cell)

Email: [Philip.Freda@cshs.org](mailto:Philip.Freda@cshs.org), [philip.freda@csmc.edu](mailto:philip.freda@csmc.edu)

Professional Website: [philipfreda.com](http://philipfreda.com)

### **EDUCATION:**

2001-2005                      BS - Administration of Justice, Pennsylvania State University, Abington, PA, United States

2012-2014                      MS - Biology, Saint Joseph's University, Department of Biology, Philadelphia, PA, United States

2014-2018                      PhD - Entomology, Focus: Ecological Genomics, Kansas State University, Department of Entomology, Manhattan, KS, United States

2019-2021                      T32 Postdoctoral Fellow - Artificial Intelligence, Machine Learning, Clinical Informatics, and Computational Biology, University of Pennsylvania, Department of Biostatistics, Epidemiology, and Informatics, Philadelphia, PA, United States

### **BOARD CERTIFICATION:**

2022-                              Responsible Conduct of Research (RCR), Cedars-Sinai Medical Center

### **PREVIOUS POSITIONS:**

08/2012-05/2014              Graduate Research Assistant/GK-12 Teaching Fellow, Department of Biology, Saint Joseph's University

08/2014-12/2018              Graduate Research Assistant, Department of Entomology, Kansas State University

05/2019-05/2021              T32 Postdoctoral Fellow, Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania

05/2021-01/2022              Postdoctoral Researcher, Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania

01/2022-12/2023              Postdoctoral Scientist, Department of Computational Biomedicine, Cedars-Sinai Medical Center

01/2024-12/2025              Project Scientist, Department of Computational Biomedicine, Cedars-Sinai Medical Center

### **PROFESSIONAL ACTIVITIES:**

#### **Other Committee Service:**

01/2013-08/2013              Member, Northeast Spotted Wing Drosophila Working Group

01/2014-05/2014              Member, Commencement Speaker Committee, Saint Joseph's University

01/2016-12/2016              Member, Lethal Weapons Event Sub-Committee, Kansas State University

08/2016-05/2017              Member, Committee on Governmental Issues, Kansas State University

08/2017-05/2018              Member, Popenoe Entomology Club - Public Relations, Kansas State University

**Professional Associations/Society Memberships:**

|          |                                                       |
|----------|-------------------------------------------------------|
| 05/2014- | Member, Graduate Honors Society, Alpha Epsilon Lambda |
| 05/2016- | Member, Golden Key Honour Society                     |
| 07/2024- | Member, K Scholar, UCLA CTSI K Scholars Society       |

**Community Service:**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/2012-        | Journal Peer Reviewer: AI in Medicine, Biological Control, BMJ Health & Care Informatics, BMJ Mental Health, Computers in Biology and Medicine, Drug and Alcohol Dependence, Entomologia Experimentalis et Applicata, Frontiers in Public Health, Genetica, Genetics, Heredity, Journal of Medical Internet Research, Journal of Neuroscience Research, Journal of Pest Science, Journal of Thermal Biology, Nature Biomedical Engineering, Nature Reviews Methods Primers, Scientific Reports, Substance Use Research and Treatment |
| 05/2013-06/2013 | Recruitment Representative, Saint Joseph's University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 12/2023-12/2023 | Grant Peer Reviewer, National Science Foundation (NSF)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 04/2026-04/2026 | Grant Peer Reviewer, NIH Center for Scientific Review, NIDA                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**MENTORING:****Undergraduate, Medical Student, and Research Intern Mentoring:**

|                 |                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/2014-05/2015 | Colin Bailey, Undergraduate, BS - Geography, Research Mentorship, Entomologist, Arizona Department of Agriculture                                                               |
| 08/2014-05/2015 | Saadia Cleve, Undergraduate, BS - Agricultural Engineering, Research Mentorship                                                                                                 |
| 08/2015-05/2016 | Ashley Helget-Wedin, Undergraduate, BS - Biology, Research Mentorship                                                                                                           |
| 08/2015-05/2016 | Oshadhi Athukorala Arachchige, Undergraduate, BS - Biology, Research Mentorship, Scientist, Sri Lanka Institute of Nanotechnology                                               |
| 08/2015-05/2016 | Mariah Brown, Undergraduate, BS - Biological Systems Engineering, Research Mentorship, Hazards Analyst, Amentum                                                                 |
| 08/2015-05/2017 | Nicholas Heter, Undergraduate, BS - Biology (Pre-med), Research Mentorship, Family Medicine Physician, UC Health                                                                |
| 08/2015-05/2017 | Jackson Alex, Undergraduate, BS - Biology, Research Mentorship                                                                                                                  |
| 08/2015-05/2017 | Adam Schieferecke, Undergraduate, BS - Microbiology, Research Mentorship, Co-Founder, Head of Research, Vivere Oncotherapies                                                    |
| 08/2016-05/2018 | Zainab Ali, Undergraduate, BS - Molecular and Cellular Biology, Genetics, Research Mentorship, Senior Global Product Manager, Software and Bioinformatics, Bio-Rad Laboratories |
| 05/2022-08/2022 | Elizabeth Zhang, Undergraduate, BS - Statistics, Research Mentorship, Health Policy Research Assistant, Center on Budget and Policy Priorities                                  |
| 05/2022-08/2022 | Tianhao Luo, Undergraduate, BA, Research Mentorship, PhD Researcher, Harvard University                                                                                         |
| 08/2024-05/2025 | Yufei Meng, Undergraduate, BS, Research Mentorship, PhD Student, New York University                                                                                            |
| 05/2025-08/2025 | Kia Kazemi-Nia, Research Intern, MS, Research Mentorship, Research Associate Data Scientist, Cedars-Sinai Medical Center                                                        |

**HONORS AND SPECIAL AWARDS:**

|      |                                                                                               |
|------|-----------------------------------------------------------------------------------------------|
| 2012 | GK-12 Teaching Fellowship, National Science Foundation, Howard Hughes Medical Institute       |
| 2012 | Graduate Assistant Scholarship, National Science Foundation, Howard Hughes Medical Institute  |
| 2014 | Biology Graduate Student Award, Saint Joseph's University                                     |
| 2014 | Travel Award, American Genetics Association                                                   |
| 2014 | Timothy R. Donahue Graduate Scholarship, Kansas State University                              |
| 2015 | Reginald H. Painter Memorial Scholarship, Kansas State University - Department of Entomology  |
| 2015 | Grants in Aid of Research (GIAR), Sigma Xi: Kansas State University                           |
| 2016 | Travel Award, Kansas State University - Department of Entomology                              |
| 2016 | Don C. Warren Genetics Scholarship, Kansas State University - Department of Entomology        |
| 2017 | Travel Award, Kansas State University - Department of Entomology                              |
| 2017 | Roger C. Smith Scholarship in Entomology, Kansas State University - Department of Entomology  |
| 2019 | Ruth L. Kirschstein Institutional National Research Service Award, University of Pennsylvania |
| 2022 | Cedars-Sinai Clinical Scholars Program, Cedars-Sinai Medical Center                           |

## **RESEARCH AWARDS AND GRANTS:**

### **Current Grants:**

8/1/2025 - 7/31/2030

*Improving the prediction of opioid outcomes via automated EHR data processing*

National Institute Drug Abuse #1K01 DA063751

Principal Investigator: **Philip J. Freda, M.S., Ph.D.**

USD 958,500.00

7/1/2026 - 6/30/2028

*Agentic AI for improving the prediction of opioid outcomes via automated EHR data processing*

National Institute Drug Abuse #1L30 DA067192

Principal Investigator: **Philip J. Freda, M.S., Ph.D.**

USD 27,194.40

### **Past Grants:**

5/15/2019 - 5/15/2021

*Ruth L. Kirschstein National T32 - Research Service Award: Institutional Research Training Grant*

National Human Genome Research Institute #T32 HG009495

Recipient: **Philip J. Freda, M.S., Ph.D.**

USD 120,000

## **INVITED LECTURES AND PRESENTATIONS:**

### **International Presentations**

1. Decoupling of physiology across metamorphosis, SICB Annual Meeting, Society for Integrative and Comparative Biology, Portland, Oregon, 1/2016

2. Identifying mechanisms of cold hardiness across metamorphosis in *Drosophila melanogaster*, International Symposium of the Environmental Physiology of Ectotherms and Plants, Estonian University of Life Sciences, Tartu, Estonia, 7/2017
3. Cluster analysis reveals socioeconomic disparities among elective spine surgery patients, PSB 2024, Pacific Symposium on Biocomputing, Kona, Hawaii, 1/2024
4. Agentic AI Methods, Systems, and Software for *in silico* Team Science, Intelligent Systems for Molecular Biology, International Society for Computational Biologists, Washington, DC, 7/2026
5. Agents in the Core: How Your Core is Using Agentic AI (Or How it Should Be), Intelligent Systems for Molecular Biology, International Society for Computational Biologists, Washington, DC, 7/2026

#### **National Presentations**

1. Genomic regions responsible for altered reproductive characteristics of *Arabidopsis thaliana* grown at elevated [CO<sub>2</sub>], Plant Biology 2012, American Society of Plant Biologists, Austin, Texas, 7/2012
2. Temporal variation at microsatellite loci in wild-caught *Drosophila simulans*, Arthropod Genomics Research Symposium, Kansas State University, Manhattan, Kansas, 6/2015
3. Phenotypic and genetic decoupling of thermal hardiness across metamorphosis, 59th Annual *Drosophila* Research Conference, Genetics Society of America, Philadelphia, Pennsylvania, 4/2018
4. A practical user's guide for ChatGPT, Health Language Processing Annual Symposium 2024, Cedars-Sinai Medical Center, West Hollywood, California, 10/2024
5. Agentic AI in Health Data Science: Building and Evaluating Autonomous Workflows, Quantitative Health Sciences 646, University of Hawai'i, Honolulu, HI (Virtual), 4/2026

#### **Regional and Extramural Local Presentations**

1. Ontogenetic constraint in the thermal physiology of *Drosophila melanogaster*, Ecological Genomics Summer Research Forum, Kansas State University, Manhattan, Kansas, 11/2015

#### **Cedars-Sinai/Institutional Presentation**

1. Automated quantitative trait locus analysis, CBM Research Day, Cedars-Sinai Medical Center - Department of Computational Biomedicine, West Hollywood, California, 8/2022
2. Predicting spine surgery outcomes and post-operative dosage from electronic health records, CBM Research Day, Cedars-Sinai Medical Center - Department of Computational Biomedicine, West Hollywood, California, 8/2022
3. Electronic health record (EHR) data challenges and solutions, CBM Science Cafe, Cedars-Sinai Medical Center - Department of Computational Biomedicine, West Hollywood, California, 9/2024
4. From data to decisions: AI in preoperative risk assessment for spinal fusion, Division of Informatics Grand Rounds, Division of Informatics, Cedars-Sinai Medical Center, West Hollywood, California, 3/2025
5. StarBASE-GP: biologically-guided automated machine learning for genotype-to-phenotype association analysis, CBM Grand Rounds, Department of Computational Biomedicine, Cedars-Sinai Medical Center, West Hollywood, California, 9/2025
6. Leveraging Automated Machine Learning to Benchmark, Deconstruct, and Compare Frailty Indices, Clinical Informatics Rotation Seminar, Division of Informatics, Cedars-Sinai Medical Center, 2/2026

#### **TEACHING ACTIVITIES:**

|                 |                                                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/2026-04/2026 | Course developed, Artificial Intelligence (HAI 100B), PhD in Health Artificial Intelligence, Cedars-Sinai Medical Center, Health Sciences University, Los Angeles, United States |
| 01/2026-04/2026 | Course taught, Artificial Intelligence (HAI 100B), PhD in Health Artificial Intelligence, Cedars-Sinai Medical Center, Health Sciences University, Los Angeles, United States    |

02/2026- Course taught, Clinical Informatics (ME-ME100.01), UCLA, Visiting Student Learning Opportunities, Clinical Resident Rotations

### **Non-Clinical Teaching**

01/2010-05/2010 Teaching Assistant, Cellular Biology, Saint Joseph's University  
 05/2010-08/2010 Teaching Intern, Noyce Scholarship Program, The School District of Philadelphia  
 08/2012-05/2014 Teaching Fellow, GeoKids LINKS GK-12 Program, Saint Joseph's University and the Wagner Free Institute of Science  
 01/2015-12/2015 Lecturer and Coordinator, R User Group, Kansas State University  
 08/2015-12/2015 Teaching Assistant and Lecturer, Insects and People (ENTOM 301), Kansas State University  
 12/2019-12/2019 Guest Lecturer, Using R in your Research, Saint Joseph's University  
 08/2023-05/2025 Invited Lecturer, Inferring Opioid Use Disorder Severity from Clinical Notes: An Annotation Study, University of Pennsylvania  
 02/2026-04/2026 Faculty Mentor, CSMC Clinical Scholars Grant Writing Workshop, Cedars-Sinai Medical Center

### **BIBLIOGRAPHY/PUBLICATIONS:**

#### **Research Papers – Peer-Reviewed (Published)**

1. Freda PJ, Braverman JM. *Drosophila suzukii*, or Spotted Wing Drosophila, Recorded in Southeastern Pennsylvania, U.S.A. Entomological News, May 01, 2013; 123, (1): (71-75). doi: 10.3157/021.123.0116
2. Leung, W, Shaffer, CD, Reed, LK, Smith, S.T., Barshop, W., Dirkes, W., [...], Freda, P.J., [...], Eglin, S.C.R. *Drosophila* Muller F Elements maintain a distinct set of genomic properties over 40 million years of evolution. G3: Genes, Genomes, Genetics, May 01, 2015; 5, (5): (719-740) doi: 10.1534/g3.114.015966. PubMed PMID: 25740935
3. Freda PJ, Alex JT, Morgan TJ, Ragland GJ. Genetic Decoupling of Thermal Hardiness across Metamorphosis in *Drosophila melanogaster*. Integrative and comparative biology, Nov 01, 2017; 57, (5): (999-1009). doi: 10.1093/icb/ix102. PubMed PMID: 29045669
4. Everman ER, Freda PJ, Brown M, Schieferecke AJ, Ragland GJ, Morgan TJ. Ovary Development and Cold Tolerance of the Invasive Pest *Drosophila suzukii* (Matsumura) in the Central Plains of Kansas, United States. Environmental entomology, Aug 01, 2018; 47, (4): (1013-1023). doi: 10.1093/ee/nvy074. PubMed PMID: 29846535
5. Freda PJ, Ali ZM, Heter N, Ragland GJ, Morgan TJ. Stage-specific genotype-by-environment interactions for cold and heat hardiness in *Drosophila melanogaster*. Heredity, Oct 01, 2019; 123, (4): (479-491). doi: 10.1038/s41437-019-0236-9. PubMed PMID: 31164731
6. Kennedy EE, Davoudi A, Hwang S, Freda PJ, Urbanowicz R, Bowles KH, Mowery DL. Identifying Barriers to Post-Acute Care Referral and Characterizing Negative Patient Preferences Among Hospitalized Older Adults Using Natural Language Processing. AMIA ... Annual Symposium proceedings. AMIA Symposium, Jan 01, 2022; 2022, : (606-615). PubMed PMID: 37128417
7. Poulsen MN, Freda PJ, Troiani V, Davoudi A, Mowery DL. Classifying Characteristics of Opioid Use Disorder From Hospital Discharge Summaries Using Natural Language Processing. Frontiers in public health, Jan 01, 2022; 10, : (850619). doi: 10.3389/fpubh.2022.850619. PubMed PMID: 35615042. **Equal First Authorship.**
8. Freda PJ, Toxopeus J, Dowle EJ, Ali ZM, Heter N, Collier RL, Sower I, Tucker JC, Morgan TJ, Ragland GJ. Transcriptomic and functional genetic evidence for distinct ecophysiological responses across complex life cycle stages. The Journal of experimental biology, Jun 01, 2022; 225, (11): (jeb244063). doi: 10.1242/jeb.244063. PubMed PMID: 35578907
9. Freda PJ, Ghosh A, Zhang E, Luo T, Chitre AS, Polesskaya O, St. Pierre CL, Gao J, Martin CD, Chen H, Garcia-Martinez AG, Wang T, Han W, Ishiwari K, Meyer P, Lamparelli A, King CP,

- Palmer AA, Li R, Moore JH. Automated quantitative trait locus analysis (AutoQTL). BIODATA MINING, Apr 10, 2023; 16, (1)doi: 10.1186/s13040-023-00331-3. PubMed PMID: 37038201
10. Orlenko A, Freda PJ, Ghosh A, Choi H, Matsumoto N, Bright TJ, Walker CT, Obafemi-Ajayi T, Moore JH. Cluster Analysis reveals Socioeconomic Disparities among Elective Spine Surgery Patients. Pacific Symposium on Biocomputing, Jan 01, 2024; 29, : (359-373). doi: 10.1142/9789811286421\_0028. PubMed PMID: 38160292. **Equal First Authorship.**
  11. Poulsen MN, Freda PJ, Troiani V, Mowery DL. Developing a Framework to Infer Opioid Use Disorder Severity From Clinical Notes to Inform Natural Language Processing Methods: Characterization Study. JMIR mental health, Jan 01, 2024; 11, : (e53366). doi: 10.2196/53366. PubMed PMID: 38224481. **Equal First Authorship.**
  12. Batista S, Madar VS, Freda PJ, Bhandary P, Ghosh A, Matsumoto N, Chitre AS, Palmer AA, Moore JH. Interaction models matter: an efficient, flexible computational framework for model-specific investigation of epistasis. BioData mining, Feb 01, 2024; 17, (1): (7). doi: 10.1186/s13040-024-00358-0. PubMed PMID: 38419006. **Equal First Authorship.**
  13. Freda PJ, Ye S, Zhang R, Moore JH, Urbanowicz RJ. Assessing the limitations of relief-based algorithms in detecting higher-order interactions. BioData mining, Oct 01, 2024; 17, (1): (37). doi: 10.1186/s13040-024-00390-0. PubMed PMID: 39354639
  14. Freda PJ, Ghosh A, Bhandary P, Matsumoto N, Chitre AS, Zhou J, Hall MA, Palmer AA, Obafemi-Ajayi T, Moore JH. PAGER: A novel genotype encoding strategy for modeling deviations from additivity in complex trait association studies. BioData mining, Oct 01, 2024; 17, (1): (41). doi: 10.1186/s13040-024-00393-x. PubMed PMID: 39394173
  15. Sha Z, Freda PJ, Bhandary P, Ghosh A, Matsumoto N, Moore JH, Hu T. Distinct network patterns emerge from Cartesian and XOR epistasis models: a comparative network science analysis. BioData mining, Dec 01, 2024; 17, (1): (61). doi: 10.1186/s13040-024-00413-w. PubMed PMID: 39732697. **Equal First Authorship.**
  16. Ghosh A, Freda PJ, Shahrestani S, Boyke AE, Orlenko A, Choi H, Matsumoto N, Obafemi-Ajayi T, Moore JH, Walker CT. Preoperative anemia is an unsuspecting driver of machine learning prediction of adverse outcomes after lumbar spinal fusion. The Spine Journal: Official Journal of the North American Spine Society, Aug 01, 2025; 25, (8): (1596-1607). doi: 10.1016/j.spinee.2025.01.031. PubMed PMID: 39892713. **Equal First Authorship.**
  17. Ghosh A, Freda PJ, Shahrestani S, Orlenko A, Scheer JK, Obafemi-Ajayi T, Moore JH, Walker CT. Leveraging automated machine learning to benchmark, deconstruct, and compare frailty indices for Predicting adverse spinal surgery outcomes. Scientific Reports, 2026. In Press. **Equal First Authorship.**

## Reviews

1. Freda PJ, Moore JH, Kranzler HR. The phenomics and genetics of addictive and affective comorbidity in opioid use disorder. Drug and Alcohol Dependence, Apr 01, 2021; 221, doi: 10.1016/j.drugalcdep.2021.108602. PubMed PMID: 33652377
2. Freda PJ, Kranzler HR, Moore JH. Novel digital approaches to the assessment of problematic opioid use. Biodata Mining, Jul 15, 2022; 15, (1): (1). doi: 10.1186/s13040-022-00301-1. PubMed PMID: 35840990
3. Li C, Mowery DL, Ma X, Yang R, Vurgun U, Hwang S, Donnelly HK, Bandhey H, Senathirajah Y, Visweswaran S, Sadhu EM, Akhtar Z, Getzen E, Freda PJ, Long Q, Becich MJ. Realizing the potential of social determinants data in EHR systems: A scoping review of approaches for screening, linkage, extraction, analysis, and interventions. Journal of clinical and translational science, Jan 01, 2024; 8, (1): (e147). doi: 10.1017/cts.2024.571. PubMed PMID: 39478779

## Preprints

1. Hernandez, JG, Ghosh, A, Freda, PJ, Meng, Y, Matsumoto, N, Moore, JH. StarBASE-GP: Biologically-Guided Automated Machine Learning for Genotype-to-Phenotype Association Analysis. arXiv, 2025.

2. Sng, LM, O'Brien, MJ, Hosking, B, Szul, P, Reguant, R, Venkatesan, M, Freda, PJ, Wang, Z, Moore, JH, Klein, AH, Kuiper, M. Novel Epistatic Interaction Between RBMS3 and CDKN2B-AS1 in Coronary Artery Disease Risk Identified by Machine Learning Tool VariantSpark. medRxiv, 2025.
3. Matsumoto, N, Choi, H, Freda, PJ, Hernandez, ME, Wang, ZP, Moore, JH. EcoXAI: Autonomous Agentic Ecosystem for Explainable Artificial Intelligence and Biomedical Discovery. bioRxiv, 2026.

## Abstracts

1. Walker CT, Freda PJ, Ghosh A, Matsumoto N, Orlenko A, Choi H, Obafemai-Ajayi T, Moore JH. Machine learning models predict outcomes of hospital re-admission, prolonged length of stay and adverse discharge disposition after elective spinal fusion surgery. AANS/CNS Joint Section of Spine and Peripheral Nerves Annual Meeting, Mar 19, 2023
2. Ghosh A, Freda PJ, Moore JH, Walker CT. Automated Novel Machine Learning Algorithms Outperform Electronic Frailty Indices in Predicting Adverse Outcomes after Lumbar Spinal Fusion. AANS/CNS Spine Section Annual Meeting, Feb 23, 2025

## Posters

1. "Computational study of flowering time genetics in *Arabidopsis thaliana* with QTL Cartographer." Sigma Xi Research Symposium. Philadelphia, Pennsylvania, 2011.
2. "Drosophila biodiversity on the campus of Saint Joseph's University." Sigma Xi Research Symposium. Philadelphia, Pennsylvania, 2012.
3. "Drosophila biodiversity on the campus of Saint Joseph's University." Sigma Xi Research Symposium. Philadelphia, Pennsylvania, 2013.
4. "Estimating phage genome sizes by pulsed-field gel electrophoresis for preliminary cluster identification." 5<sup>th</sup> Annual SEA-PHAGES Symposium. Janelia Farm Research Campus, Ashburn, Virginia, 2013.
5. "Temporal variation at microsatellite loci in wild-caught *Drosophila simulans*." Sigma Xi Research Symposium. Philadelphia, Pennsylvania, 2014.
6. "Temporal variation at microsatellite loci in wild-caught *Drosophila simulans*." Sigma Xi Research Symposium. Manhattan, Kansas, 2015.
7. "Evolution of complex life cycles: Is performance constrained across metamorphosis?" 13<sup>th</sup> Annual Ecological Genomics Symposium Poster Session. Manhattan, Kansas, 2015.
8. "Decoupling of physiology across metamorphosis." Kansas State University Department of Entomology Seminar Series. Manhattan, Kansas, 2015.
9. "Ontogenetic constraint in the thermal physiology of *Drosophila melanogaster*: A genomic assessment of the adaptive decoupling hypothesis." Kansas State University Department of Biology Ecological Genomics Summer Research Forum. Manhattan, Kansas, 2015.
10. "Phenotypic plasticity promotes persistence of an invasive pest following environmental stress." Sigma Xi Research Symposium. Manhattan, Kansas, 2016.
11. "Decoupling of physiology across metamorphosis." Kansas State University Department of Entomology Seminar Series. Manhattan, Kansas, 2016.
12. "Phenotypic and genetic decoupling of thermal hardiness across metamorphosis." 59<sup>th</sup> Annual *Drosophila* Research Conference Poster Session. Philadelphia, Pennsylvania, 2018.
13. "Automated quantitative trait locus analysis (AutoQTL)." Cedars-Sinai Research Day Poster Session. West Hollywood, California, 2022.
14. "Predicting spine surgery outcomes and post-operative opioid dosage from electronic health records." Cedars-Sinai Research Day Poster Session. West Hollywood, California, 2022.
15. "Cluster analysis reveals socioeconomic disparities among elective spine surgery patients." Pacific Symposium for Biocomputing Poster Session. Kona, Hawaii, 2024.
16. "An endogenous opioid system in the Pacific Hagfish, a vertebrate-invertebrate species, analogous to the Mammalian opioid system." Society of Neuroscience Conference. San Diego, California, 2025.

## EXTRACIRRICULAR ACTIVITIES:

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| 01/2012-12/2014 | Columnist, Practical Science with Phil Freda, Patch.com, Upper Moreland-Willow Grove, PA |
| 08/2014-08/2017 | Team Manager, Intramural Softball, Kansas State University                               |